This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

A Q-Learning-Based Hyper-Heuristic for Capacitated Electric Vehicle Routing Problem

Jian Lin, *Member, IEEE*, Xintao Wang, Rui Niu, and Yifan He, *Member, IEEE*

***Abstract*—The capacitated electric vehicle routing problem (CEVRP) requires a set of shortest delivery routes for capacitated electric vehicles from a depot to multiple customer locations. In this paper, a Q-learning-based hyper-heuristic (QHH) is proposed to solve the CEVRP. First, a novel initialization method is proposed to generate capacity-feasible solutions with potentially shorter routes. Second, a repair method is devised to insert proper charging stations into routes, generate feasible solutions, optimize visits to charging stations, and subsequently improve the solutions. Then, a diverse set of six common low-level heuristics (LLHs) and several local search methods are employed as fundamental components of the QHH algorithm. Meanwhile, Q-learning serves as a guiding high-level heuristic, dynamically selecting the appropriate LLHs in response to the evolutionary process. Additionally, the Design of Experiments methodology is employed to investigate the performance of various parameter settings. Finally, the proposed QHH is evaluated on a large set of 60 CEVRP instances, demonstrating its superiority over four state-of-the-art algorithms.**

***Index Terms*—Capacitated, electric vehicle, hyper-heuristic, Q-learning, vehicle routing problem.**

I. Introduction

TRANSPORTATION has emerged as a primary contributor to carbon emissions for decades [1]. The recent development of electric vehicles (EVs) has provided a promising alternative to conventional fossil-fueled vehicles. Studies have indicated that the transition from conventional vehicles to EVs holds substantial potential for carbon emission reduction [2], [3]. Consequently, numerous logistics enterprises have initiated decarbonization in their delivery business, opting to substitute conventional delivery vehicles with EVs. For example, Amazon transported more than 680 million packages via EVs in 2023, which is part of their ongoing initiative to achieve "Net Zero Carbon" emissions by 2040 [4]. According to the survey conducted by Golden et al. [2], a notable trend has emerged in the field of green routing. Specifically, from 2017 to 2022, 12 survey articles were published focusing on this topic. This number accounts for approximately 15% of all survey articles published during this period and highlights the growing interest and significance of green routing within the broader context of the vehicle routing problem (VRP) research.

VRP aims to optimize the routing of vehicles from a central depot to various customer locations to ensure efficient delivery operations [2], [5], [6], [7], [8]. Moreover, VRP could serve as the fundamental model of various real-world applications, even when the concept of "vehicles" is not directly involved. For example, Öztürk et. al proposed a path-finding and task allocation algorithm for multi-robotic scheduling problems [7] and later extended the method to solve the autonomous runway foreign object debris clearance problem [8]. These problems can be formulated as VRP as well, where each robot is recognized as a "vehicle".

In the past decades, the increased volume of delivery parcels has made it necessary to consider the capacity limitation of vehicles, resulting in a notable variant known as the capacitated VRP (CVRP) [9], [10]. The capacity constraint turns the problem into a more realistic but challenging task [9]. Meanwhile, EVs have gained widespread usage for their eco-friendly and energy-efficient characteristics in recent years. However, these EVs are restricted by their battery capacity, resulting in a limited distance within a single trip. Therefore, efficient routing that includes scheduled visits to charging stations becomes imperative for the EV-based VRP variant, referred to as the electric VRP (EVRP) [11]. Similarly, the capacitated electric VRP (CEVRP) [12] can be regarded as a variant of CVRP [13] wherein all delivery vehicles are EVs. The primary distinction between the CEVRP and the CVRP lies in the inclusion of battery constraints. Specifically, the scheduled delivery route must ensure sufficient battery for traversing between any adjacent pair of customer locations along the route. To meet this requirement, scheduled visits to charging stations are necessary. In this way, the battery constraint significantly expands the search space of the problem. Furthermore, there is a trade-off between satisfying the battery constraint and minimizing the total distance, as visiting a charging station often results in extended route lengths. This interplay renders CEVRP a challenging task. Beyond its complexity, this NP-hard problem finds applications in various real-world scenarios such as delivery and logistics operations [14], [15].

Various algorithms have been proposed to tackle the VRP and its variants, including exact algorithms [16], [17], [18], heuristics [19], [20], and meta-heuristics. Exact algorithms excel in solving small-scale problems; however, their computational time escalates significantly when solving larger

Received 28 May 2024; revised 22 December 2024 and 25 May 2025; accepted 28 July 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 61973267 and Grant 61503331 and in part by Zhejiang Provincial Natural Science Foundation of China under Grant LY24F030008. The Associate Editor for this article was H. Jula. *(Corresponding author: Yifan He.)*

The authors are with the School of Information Technology and Artificial Intelligence, Zhejiang University of Finance and Economics, Hangzhou 310018, China (e-mail: linjian1001@126.com; 18757546201@163.com; 15968102014@163.com; yifanhe@zufe.edu.cn).

This article has supplementary downloadable material available at https://doi.org/10.1109/TITS.2025.3594393, provided by the authors.

Digital Object Identifier 10.1109/TITS.2025.3594393

instances. Heuristics offer an efficient approach to obtain suboptimal solutions within a short period [19], [20], [21], [22]; however, many times these heuristics can fall into local optima. Meta-heuristics integrate random and local search strategies and typically exhibit strong global search capabilities. This characteristic enhances their performance in addressing challenging optimization problems.

CEVRP [12] stands as one of the most recent and difficult variants of the VRP. A large number of solution methodologies for CEVRP are rooted in meta-heuristic approaches. Woller et al. [23] introduced a method grounded in greedy randomized adaptive search procedure to tackle this problem. Jia et al. [14] proposed a bi-level ant colony optimization algorithm, which divides the CEVRP into two levels of subproblems. The upper-level adheres to capacity constraints and is addressed by an ant colony system, whereas the lower-level focuses on battery constraints and is solved through a heuristic method. Wang et al. [24] proposed using GA to construct the final solution and incorporating elite strategies into GA to augment the global search capabilities. Leite et al. [25] combined variable neighborhood descent for the lower-level station allocation problem with improved ant colony optimization for the upper-level VRP. This method was further developed by Jia et al. [26] by incorporating a confidence-based selection method to select promising customer sequences to conduct local search and lower-level optimization. Hien et al. [19] proposed a hybrid evolutionary algorithm based on greedy search to solve the CEVRP. Bui and Mai [27] presented an improvement learning approach integrated with imitation learning, leveraging classical heuristics to guide the policy model towards superior solutions. Chen et al. [28] proposed a threshold acceptance based multilayer search to solve this problem. Pustilnik and Borrelli [29] introduced a method for the robust energy CVRP. Though meta-heuristics have demonstrated promising results on CEVRP, their performance can vary across different problem/instance landscapes, as posited by the renowned No Free Lunch theorem [30]. Moreover, designing efficient meta-heuristics for a specific problem/instance is a non-trivial task and requires domain knowledge, algorithm design techniques, and numerous efforts of experiments.

Hence, it is particularly necessary to devise an algorithm capable of adapting to diverse instances automatically. For this very reason, the concept of hyper-heuristic (HH) has been proposed. HH represents a set of algorithms that utilize a high-level heuristic (HLH) to manage or manipulate a collection of problem-specific low-level heuristics (LLHs) based on the feedback received from the optimization process [31], [32], [33]. HH facilitates the composition of new (meta-)heuristics tailored to different instances automatically. In contrast to (meta-)heuristics, HH operates on a space of LLH combinations, making it more universally applicable [34]. HH has proven to be a powerful solver on various combinatorial problems, such as job-shop scheduling [35], traveling salesman problem [36], search-based software engineering [37], etc.

Various methods have been applied for the high-level selection strategy of HH, such as GA [38], tabu search [39], ant colony optimization [36], and reinforcement learning (RL) [40]. Particularly, RL-based HHs have gained recent attention due to their ability to leverage historical information accumulated during the problem-solving process through online learning. By adapting the selection and utilization of LLHs, RL-based HHs enhance algorithm performance and applicability [41], [42]. Q-learning is a widely used RL algorithm, which determines the optimal action by learning the action-value function (Q-function) [43]. This simple yet powerful RL algorithm has been applied in various fields, such as autonomous mobile robots [44], industrial process control [45], etc.

In this paper, we introduce a Q-learning-based HH (QHH) to address the CEVRP, filling a research gap in this domain. We select six effective LLHs that are frequently used in the literature of VRP. Meanwhile, Q-learning is employed to manage these LLHs. To overcome challenges associated with conventional reward functions, such as feedback insensitivity, reward sparsity, and susceptibility to local optima [46], we employ an adaptive reward function that adjusts reward values based on the evolutionary process to ensure sustained and effective learning. Following the execution of LLHs, a 2-opt local search is executed to further refine the solution. Moreover, we adopt threshold accepting (TA) [47] as the solution acceptance strategy to enhance exploration. To the best of our knowledge, there is no published work proposing QHH to solve the CEVRP. We validate the superiority of the QHH over four state-of-the-art algorithms on 60 CEVRP instances. The main contributions of this study are summarized as follows.

1) A novel initialization strategy is proposed to employ potentially fewer vehicles and generate potentially shorter capacity-feasible routes for the CEVRP.
2) A new repair strategy using roulette wheel charging station selection is introduced to convert the capacity-feasible routes into electricity-feasible routes.
3) Q-learning with problem-specific state representation and adaptive reward function is applied as the HLH to realize the online management of a set of LLHs for the first time on the CEVRP.
4) Better optimization performance with statistical significance is achieved compared to four state-of-the-art algorithms across a wide range of benchmark instances.

The remainder of this paper is organized as follows. Section II introduces the CEVRP and its mathematical formulation. Section III provides an overview of the Q-learning algorithm. Section IV delves into the detailed design of our QHH, discussing its key components and functions. Section V showcases experimental results on CEVRP instances, comparing our QHH with state-of-the-art algorithms. Section VI summarizes the main research content and outlines several future directions.

II. The Capacitated Electric Vehicle Routing Problem

A. Problem Description

Given a fleet of homogeneous EVs, CEVRP [12] aims to determine the optimal delivery routes from a starting node

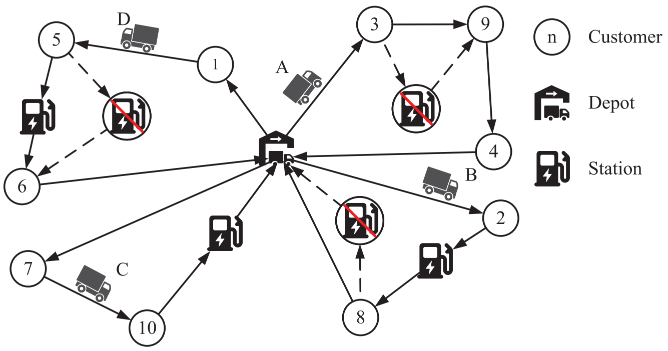


Fig. 1. An example of CEVRP.

to multiple customers, while considering load capacity and battery capacity. This optimal route ensures that each EV can complete the delivery tasks and return to the starting node. In the CEVRP, EVs are allowed to recharge at charging stations and must maintain sufficient battery capacity to reach the next node at each stage of the delivery process. This modification significantly expands the search space and introduces a challenging constraint.

The detailed description of the CEVRP is as follows. With a distribution center serving as both the starting and destination node, the objective is to find the optimal solution (i.e., a set of delivery routes with minimum traveling distance) that facilitates the delivery of goods to a predefined set of customer nodes. Moreover, this optimal solution should satisfy the following rules.

- For each route, the total demand of the customers must not exceed the maximum load capacity of the EVs.
- Each EV starts from the distribution center in a fully charged state.
- Each customer can only be served by one EV at a time.
- The battery consumption of EVs is linear, with a defined battery consumption rate.
- EVs leave the charging station fully charged, and each charging station can be visited multiple times.

Fig. 1 illustrates the delivery and charging scenarios related to the CEVRP. Some vehicles equipped with sufficient battery capacity (e.g., vehicle A) can cover the entire delivery route, while others (e.g., vehicles B, C, and D) require interim charging to reach the next customer node successfully. Hence, the choice of charging stations and the timing of charging significantly impact route length and delivery efficiency in CEVRP.

B. Mathematical Formulation of the Cevrp

The CEVRP [12] can be modeled using an undirected graph $G = (V, E)$. The node set V consists of a distribution center p, a set of n customers $I = \{i_1, \ldots, i_n\}$, and m charging stations $C = \{c_1, \ldots, c_n\}$. $E = \{(i, j) \mid i, j \in V, i \neq j\}$ is the set of edges, each of which is associated with a weight d_{ij} representing the distance between i and j. The mathematical formulation of the CEVRP [12] is as follows. A list of notations used in this formulation is provided in Table I.

$$\min \Sigma_{i,j \in V, i \neq j} d_{ij} x_{ij} \tag{1}$$

TABLE I
NOTATIONS IN THE MATHEMATICAL FORMULATION OF CEVRP

Variable	Definition
x_{ij}	Whether there is a vehicle from node i to node j
d_{ij}	Distance from node i to node j
w_i	Remaining load capacity at node i
l_i	Demand of the customer at node i
W	Maximum load capacity of each vehicle
q_i	Remaining battery capacity at node i
h	Battery consumption rate
Q	Maximum battery capacity of each vehicle

$$\text{s.t. } x_{ij} \in \{0, 1\}, \quad \forall i, j \in V, i \neq j \tag{2}$$
$$\Sigma_{j \in V, j \neq i} x_{ij} = 1, \quad \forall i \in I \tag{3}$$
$$\Sigma_{j \in V, j \neq i} x_{ij} \leq 1, \quad \forall i \in C \tag{4}$$
$$\Sigma_{j \in V, j \neq i} x_{ij} - \Sigma_{j \in V, j \neq i} x_{ji} = 0, \quad \forall i \in V \tag{5}$$
$$w_j \leq w_i - l_i x_{ij} + W(1 - x_{ij}), \quad \forall i, j \in V, i \neq j \tag{6}$$
$$0 \leq w_i \leq W, \quad \forall i \in V \tag{7}$$
$$q_j \leq q_i - h d_{ij} x_{ij} + Q(1 - x_{ij}), \quad \forall i, j \in V, i \neq j \tag{8}$$
$$0 \leq q_i \leq Q, \quad \forall i \in V \tag{9}$$
$$q_j \leq Q - h d_{ij} x_{ij}, \quad \forall i \in C \cup \{p\}, j \in V, i \neq j \tag{10}$$

Eqs. (1) and (2) represent the objective function and decision variables used in CEVRP. Eq. (3) ensures the connectivity between two nodes. Eq. (4) ensures that each charging station serves only one EV at a time. Eq. (5) equates the number of paths entering and leaving the node. Eqs. (6) and (7) indicate that the customer demand at any node is a non-negative value not exceeding the maximum load capacity W. Eqs. (8)- (10) indicate that the battery capacity of EVs at any node is between 0 and the maximum battery capacity Q.

III. Background of Q-Learning Algorithm

RL has emerged as a prominent field within machine learning, characterized by the ability of agents to optimize strategies through iterative trial-and-error interactions within a defined environment [48]. Among RL methods, Q-learning [48] is widely adopted for its capacity to identify optimal actions in specific states to maximize cumulative rewards. This is achieved by iteratively refining a value function (i.e., Q-values) based on experience.

The core elements of Q-learning are states, actions, rewards, and Q-values. States represent the conditions an agent encounters, actions are the available decisions in each state, rewards are the immediate feedback post-action, and Q-values represent the expected utility of an action in a state under an optimal policy. These Q-values are stored in a structured Q-table and iteratively updated using Eq. (11).

$$\begin{aligned} Q_{t+1}(s_t, a_t) = Q_t(s_t, a_t) + \alpha[r_t \\ + \gamma \arg\max Q(s_{t+1}, a) - Q_t(s_t, a_t)] \end{aligned} \tag{11}$$

s_t represents the state at time t, a_t represents the action at time t and r_t denotes the immediate reward obtained when the agent takes action a_t at state s_t. $\alpha \in [0, 1]$ is the learning rate, determining the degree to which the agent accepts new information. $\gamma \in [0, 1]$ is the discount factor, indicating the importance of future rewards. $Q_t(s_t, a_t)$ represents the Q-value at time t.

Typically, the ε-greedy strategy is commonly used for action selection: it selects the action with the highest Q-value with probability 1-ε and a random action with probability ε. This balances exploitation of known high-Q actions and exploration of other actions. The iterative application of this strategy and the Bellman update rule allows Q-learning to converge to optimal Q-values, enabling effective decision-making. The main procedure of Q-learning is as follows:

Step 1: Initialize the Q-table.

Step 2: Select the action based on the current state using the ε-greedy strategy.

Step 3: Execute the action, compute the reward, and determine the next state.

Step 4: Update the Q-value using Eq. (11).

Step 5: Repeat Steps 2–4 until the stopping criteria are met.

Q-learning is a simple yet efficient model-free RL algorithm, and therefore can operate effectively in various complex environments. Identifying the optimal LLHs during the optimization is definitely a non-trivial task. The optimal LLHs can vary depending on the specific problem instances and even different optimization stages when solving the same instance. Moreover, Q-learning can be implemented as an online learning algorithm, which simultaneously makes decisions and learns from the environment. This property facilitates the direct utilization of the algorithm without the need for preliminary training. These advantages motivate us to propose an online hyper-heuristic algorithm based on Q-learning. The settings of the Q-learning in this research are detailed in Sections IV-D- IV-H.

IV. QHH FOR CEVRP

A. Encoding and Decoding Schemes

In the QHH, we adopt a frequently used encoding and decoding scheme in the literature. Delivery routes are encoded as multiple lists, where each list represents a specific route. The integers in the route denote the customers and charging stations to be visited. The order to visit these nodes is the same as the order they appear in the list. Fig. 2 illustrates the encoding and decoding schemes with an example solution consisting of three routes, (0, 1, 5, 9, 6, 0), (0, 8, 2, 3, 10, 0), and (0, 11, 7, 12, 4, 0). Node 0 represents the central depot, nodes 1 to 8 are customer nodes, and nodes 9 to 12 are charging stations.

B. Initialization Strategy

In the CEVRP, routes that satisfy the capacity constraints are usually referred to as capacity-feasible routes (c-routes), and those that also satisfy battery constraints are named as electricity-feasible routes (e-routes). We denote c-routes by φ and e-routes by Ψ, respectively. A novel initialization strategy, as outlined in Algorithm 1, is proposed to generate a set of c-routes from the customer set I. Specifically, given a shuffled customer list I', we visit each node i in the list and insert it into an empty route. If the capacity constraint is violated after inserting node i, we revoke the operation and proceed to the next node. After visiting all nodes, a c-route is constructed. However, some nodes may remain in I' at the moment. We then repeat the previous steps until I' is empty and create multiple routes. Compared with the common initialization method [9], this novel initialization requires fewer vehicles and generates potentially shorter c-routes.

Algorithm 1 Solution Initialization

Input: maximum load capacity W and customer set I
Output: initialized solution x_0

c-route set $R \leftarrow \emptyset$, e-route set $R' \leftarrow \emptyset$, accumulated weight acw $\leftarrow 0$;
$I' \leftarrow$ randomly shuffle I;
while $|I'| > 0$ **do**
  $\varphi \leftarrow \emptyset$, acw $\leftarrow 0$;
  for $i \in \{1, 2, \ldots, |I'|\}$ **do**
    if acw $+ w_i \leq W$ **then**
      $\varphi \leftarrow \varphi \cup \{I'_i\}$;
      $I' \leftarrow I' \setminus \{I'_i\}$;
      acw $\leftarrow$ acw $+ w_i$;
  add distribution center p on both sides of φ;
  $R \leftarrow R \cup \{\varphi\}$;
$R \leftarrow$ execute 2-opt on R;
$R' \leftarrow$ execute GIRR on R;
$x_0 \leftarrow$ construct initial solution with R and R';
return x_0

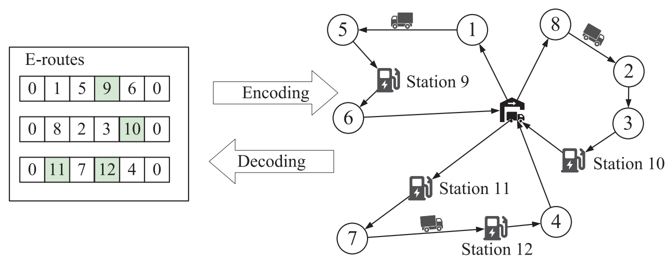


Fig. 2. An example of solution encoding and decoding schemes.

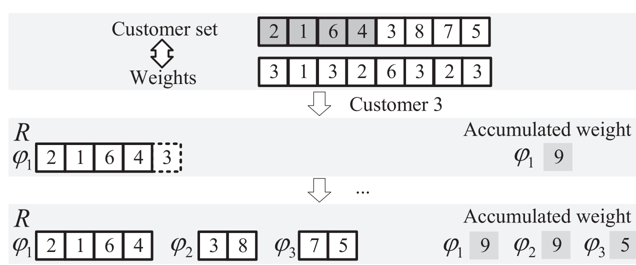


Fig. 3. An example of initialization steps.

As depicted in Fig. 3, from a shuffled customer set (2, 1, 6, 4, 3, 8, 7, 5), the customer nodes (2, 1, 6, 4) have been selected and assembled into a c-route φ_1. Given a maximum load capacity W of 10, the insertion of customer node 3 is not allowed in φ_1, as it exceeds the capacity limitation. Similarly, customer nodes 8, 7, and 5 are also prohibited due to the same constraint. Consequently, no additional customer nodes can be inserted into φ_1. Following this, we construct another c-route

φ_2 and select the remaining customer nodes from the set I', which are (3, 8, 7, 5). Repeating the previous steps, we can obtain a set of c-routes R as in Fig. 3.

C. Repair Operator

The routes generated by the procedures in the previous steps may not satisfy the battery constraints. Hence, we perform a repair on the initial sets R after initialization. The repair strategy is designed to insert an optimal set of charging stations into the c-routes to ensure compliance with the battery constraint.

Obviously, enumerating all possible charging station sets is time-consuming. For this reason, we propose an effective insertion strategy named greedy insertion and roulette selection-based remove (GIRR) upon the work of Jia et al. [14]. The procedure of GIRR can be divided into two steps. First, the optimal charging station is inserted between each pair of nodes in every c-route. Then, the charging stations are consecutively removed based on the improvement achieved through the roulette selection strategy.

Algorithm 2 GIRR

Input: c-route φ, customer set I, and charging station set E

Output: repaired e-route Ψ

$\Psi \leftarrow \varphi$;
$\Psi \leftarrow$ insert the best charging stations between every customer pair;
cnt $\leftarrow$ the amount of charging stations in Ψ;
while cnt > 0 **do**
  $D \leftarrow \emptyset$;
  for $i \in \{1, 2, \ldots, |\Psi| - 2\}$ **do**
    if $\Psi_i \in C$ *and* Ψ *is electric-feasible after removing* Ψ_i **then**
      $D \leftarrow D \cup \{(\Psi, d_{\Psi_{i-1},\Psi_i} + d_{\Psi_i,\Psi_{i+1}} - d_{\Psi_{i-1},\Psi_{i+1}})\}$;
  $\Psi \leftarrow$ remove one charging station based on the value in D with roulette wheel selection strategy;
  cnt $\leftarrow |D|$
return Ψ

In an optimal solution, the charging station positioned between two customers is intuitively considered the most suitable. However, only relying on this heuristic, which removes the worst charging station one by one, cannot guarantee finding the optimal solution. Hence, we have adopted the roulette wheel selection strategy to select stations for deletion. This modification introduces additional randomness to ensure a more rigorous and comprehensive exploration of potential e-routes compared with the previous repair operator [14]. The probability P_{Ψ_u} to select a charging station u in an e-route Ψ can be obtained as in Eqs. (12) and (13), where u and v are the indices of the nodes in one route, and d represents the distance between a pair of nodes. The pseudocode of GIRR is provided in Algorithm 2. Fig. 4 provides an example of GIRR. First, the optimal charging station is inserted between each pair of nodes in every c-route. Then, the charging stations are consecutively removed based on the improvement achieved through the roulette selection strategy.

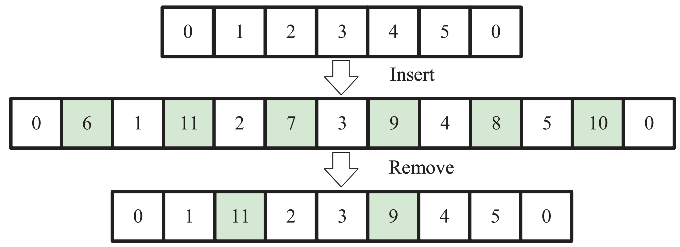


Fig. 4. An example of GIRR.

$$P_{\Psi_u} = \frac{d_{\Psi_{u-1},\Psi_u} + d_{\Psi_u,\Psi_{u+1}}}{D_\Psi} \tag{12}$$

$$D_\Psi = \Sigma_{v=1}^{|\Psi|}(d_{\Psi_{v-1},\Psi_v} + d_{\Psi_v,\Psi_{v+1}}), \quad \forall \Psi_v \in C \tag{13}$$

Note that for a c-route, the reversed route after repair steps may result in a different solution. Therefore, after repairing a c-route, we also repair its reversed route and select the one with a smaller total length of e-routes.

D. Action Representation

According to the characteristics of CEVRP, we have designed a set of LLHs in Fig. 5 to minimize the length of the c-route (i.e., the route without charging stations). This design assumes that when the length of the c-route decreases, the route length after repair is more likely to decrease. The designed LLHs fall into two categories: internal adjustment and external adjustment. Internal adjustment involves changing the order of nodes within a single c-route, while external adjustment can modify multiple c-routes simultaneously.

Internal Point Swap (IPS): Randomly select a c-route and swap two random customers.

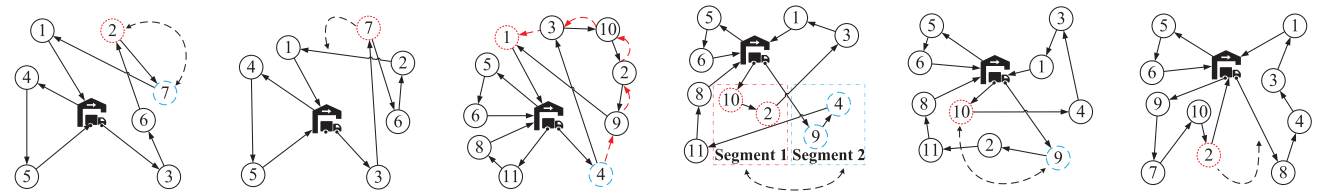
Internal Point Insertion (IPI): Randomly select a c-route, select a node at random and insert it before a random position.

Internal 2-Opt (IO): Randomly select a c-route, select two nodes $I_{i,a}$ and $I_{i,b}$ at random ($a \le b$) and reverse the segment between $I_{i,a}$ and $I_{i,b}$ ($I_{i,a}$ and $I_{i,b}$ included).

External Segment Swap (ESS): Randomly select two segments $l_{i,a,b}$ and $l_{j,u,v}$ from two different c-routes φ_i and φ_j, and swap the selected segments.

External Point Swap (EPS): Randomly select a node $I_{i,a}$ from a c-route φ_i, then select one of the closest nodes $I_{j,b}$ from a different c-route φ_j, swap the selected nodes. Specifically, we identify up to 60 nearest customer nodes from distinct c-routes. Then, we compute the gain of LLHs when operating with a node as the difference in total route length before and after its application $\Delta L = L_{pre} - L_{post}$. A roulette selection step ensures that customer nodes with larger gains are more likely to be selected to explore potentially high-quality e-routes.

External Point Insertion (EPI): Randomly select a node $I_{i,a}$ from φ_i, then select the optimal position of a different c-route φ_j, insert $I_{i,a}$ before the selected position. This optimal position is obtained by exhaustively trying all possible positions in that route. EPI first identifies the optimal insertion position in a c-route that is different from the current one, and the gain of LLHs when operating with this optimal position is



(a) IPS (b) IPI (c) IO (d) ESS (e) EPS (f) EPI

Fig. 5. The six low-level heuristic operators.

recorded. A roulette selection step is then performed to choose a c-route φ_j based on the gain of the LLH.

Algorithm 3 Action Execution

Input: LLH llh_t, threshold discount coefficient λ, solution x_t with its c-route length L_t
Output: solution x_{t+1}, best solution $x_{b,t}$
$z \leftarrow L_t \cdot \lambda$, cnt $\leftarrow 0$, $f_{b,t} \leftarrow \infty$;
while cnt $< \mu$ **do**
  for $i \in \{1, 2, \ldots, \xi\}$ **do**
    $x_t' \leftarrow \mathrm{llh}_t(x_t)$;
    if rand() $< P_{\mathrm{opt}}$ **then**
      $x_t' \leftarrow$ execute 2-opt on the c-routes in x_t';
    $L_t' \leftarrow$ evaluate the c-routes in x_t';
    if $L_t' - L_t \leq z$ **then**
      $x_t \leftarrow x_t'$, $L_t \leftarrow L_t'$;
  $x_t \leftarrow$ repair x_t with GIRR;
  $f_t \leftarrow$ evaluate the e-routes in x_t;
  if $f_t < f_{b,t}$ **then**
    $x_{b,t} \leftarrow x_t$, $f_{b,t} \leftarrow f_t$;
  cnt $\leftarrow$ cnt $+ 1$;
$x_{t+1} \leftarrow x_t$;
return x_{t+1}, $x_{b,t}$

To satisfy the capacity constraint of the problem, we repeat the LLHs 50 times if the new solution violates the constraint. If the constraint remains unsatisfied, we simply revert the solution to its parent. Moreover, a longer c-route may occasionally result in a shorter e-route. To circumvent this contradiction between the c-route and the repaired e-route, we employ a probabilistic search method to enhance the search capability. The pseudocode of the action execution is provided in Algorithm 3.

In Algorithm 3, QHH perturbs the c-route with a LLH ξ times and updates the c-route with improved candidates. Subsequently, the algorithm inserts charging stations using GIRR and generates a feasible e-route. These two steps are repeated for μ iterations, and the solution is refined by incorporating shorter feasible e-routes discovered during this process. This entire sequence constitutes one execution of an action defined by a specific LLH. The operation of repair after executing a LLH makes it possible to search for a better solution in the process of a_t. λ represents the threshold discount coefficient. P_{opt} represents the probability of performing 2-opt local search [49]. However, the optimal length of e-routes after being repaired is not always derived from the shortest c-routes. Moreover, 2-opt may find a local optimal solution based on the order of the nodes it visited. Hence, we propose an effective method by adopting a small probability to employ 2-opt after an internal adjustment LLH has been executed. The pseudocode of the 2-opt local search can be found in part I of the supplementary materials.

Algorithm 4 Threshold Control

Input: parameter λ, η_0, and solution x_t with the length of its c-routes L_t
Output: adjusted value of λ
1 **if** $t \mod \nu = 0$ **then**
2   **if** $L_t \cdot \lambda > \mathrm{BL}$ **then**
3     $\lambda \leftarrow \lambda \cdot \kappa$, $\eta \leftarrow \eta_0$;
4   **else**
5   **if** $\eta < 1$ **then**
6       $\lambda \leftarrow \lambda + \delta / L_t$;
7       $\delta \leftarrow \delta + b$, $\eta \leftarrow \eta_0$;
8   **else**
9       $\eta \leftarrow \eta - 1$;

Algorithm 4 provides the algorithmic procedure of the threshold control technique used in our QHH. This threshold determines the probability to accept a candidate when its fitness is worse than the parent. This setting allows a degree of exploration and thus helps the algorithm to escape from local optima. In Algorithm 4, λ represents the threshold discount coefficient, which enables QHH to consider inferior solutions, thereby augmenting global exploration. Moreover, a threshold controlling strategy is further proposed to adjust the value of λ based on the evolutionary process in Algorithm 4, which resembles an enhanced version of simulated annealing. Specifically, λ decreases by generations until it reaches a value determined by a pre-defined parameter BL and the length of the route in Line 2 of Algorithm 4 (BL is set to 1 in later experiments). As λ governs the likelihood of accepting inferior solutions, QHH demonstrates reduced exploration during this phase. However, if λ becomes smaller than this value for $\eta \cdot \nu$ consecutive generations, QHH runs a re-heating process to increase the value of λ and restart exploration.

E. State Representation

A proper representation of states is crucial for Q-learning. In the proposed QHH, we adopt a state representation based on fitness [50], where the improvement in fitness is discretized to represent different states. In Eq. (14), P_t represents the state of an agent at the t-th generation based on the fitness improvement of the solution, f_t represents the fitness of the solution at the t-th generation, and f_{t-1} represents the fitness at the previous generation. In QHH, the proportion P_t is carefully segmented into five aggregate states, which include $(-\infty, -3.5)$, $[-3.5, -1)$, $[-1, 1)$, $[1, 3.5)$, and $[3.5, \infty)$. These segments serve as boundary constraints to prevent any states from falling outside the designated ranges.

$$P_t = -\frac{f_t - f_{t-1}}{f_{t-1}} \tag{14}$$

F. Reward Function

After executing a chosen action, assigning appropriate rewards, and updating them in the Q-table can either reinforce or penalize the agent from taking the same action when encountering the same state again in the future. Hence, an appropriate reward resulting from a single action can significantly impact the performance of QHH. However, it is easy for the solutions that improved in the early period to receive incorrect rewards. To address this issue, we introduce an adaptive reward strategy in QHH. The reward r_t at timestep t is defined as in Eq. (15).

$$r_t = \max\left\{M, \frac{t}{T}\right\} \cdot \left(\frac{f_{t-1} - f_{b,t}}{f_{t-1}} + \beta\right) \tag{15}$$

In Eq. (15), M represents the minimum value of the reward proportionality coefficient, where $f_{b,t}$ represents the total length of e-routes of $x_{b,t}$ after executing the action a_t, t and T represent the current and the maximum execution time, respectively. Therefore, the proposed reward is mainly determined by two factors: 1) fitness improvement after executing the selected action realized by the term $f_{t-1} - f_{b,t}$ and 2) the current generation realized by the term t/T. Obviously, a larger improvement in fitness leads to a larger reward. Moreover, the same amount of improvement in the later generations can lead to a larger reward as well. This setting may help to address the sparsity of reward in the ending stage of the optimization. It is worth noting that a user-defined bias parameter β is introduced in Eq. (15). Usually, a larger value of β will be used when the solution is improved and a smaller value otherwise (notice that β can be a negative value). This reward design grants the action an opportunity for trial and error to enhance the global search ability.

G. Action Selection Strategy

We apply the ε-greedy strategy to empower the agent to maintain the balance between exploitation and exploration. With this strategy, the algorithm holds a probability of 1-ε to select the action with the highest Q-value (i.e., exploit the knowledge gained so far in the current Q-table) and a probability of ε to explore all possible actions at random (i.e., explore all possible actions and obtain more knowledge). The pseudocode of this strategy is provided in part I of the supplementary materials. The parameter ε plays a crucial role in determining the balance between exploitation and exploration. Usually, a larger value of ε encourages more exploration, while a smaller value shifts the focus towards exploitation. In the later experiments, this parameter is calibrated specifically for the CEVRP.

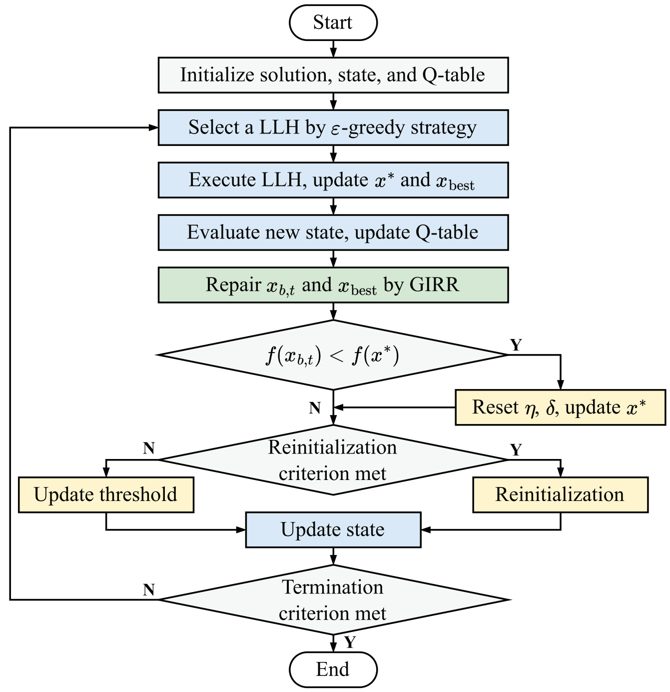


Fig. 6. Procedure of the proposed QHH.

H. Procedure of Qhh

The procedure of the QHH is detailed in Fig. 6 and can be described as follows.

Step 1: Initialize the graph G, the maximum battery capacity Q, the maximum load capacity W, the set of charging stations C, and the set of customer nodes I. Initialize a Q-table with the same initial action values Q_{init}, generate a feasible solution, and set the initial state's fitness to infinity.

Step 2: Choose an action a_t based on the ε-greedy strategy.

Step 3: Execute the action a_t. If a better solution is found, update the best solution x_{best} and staged best solution x^*. Denote the best candidate in this step as $x_{b,t}$.

Step 4: Update Q-table using Eqs. (14) and (15) based on $x_{b,t}$.

Step 5: Repair the new solution x_{t+1} and x_{best}. Update x_{best} if a better solution is found.

Step 6: If $x_{b,t}$ is better than x^*, reset the parameter η and δ. Update x^* to $x_{b,t}$.

Step 7: If x^* has not improved for Ω generations, reinitialize the solution and assign it to x^*. Otherwise, adjust the threshold control to update λ.

Step 8: Update state s_{t+1} with x_{t+1}.

Step 9: Repeat Steps 2–8 if the stopping criteria are not met.

Based on the above steps, one may notice that the proposed QHH facilitates an online-learning procedure and only employs a single-point search process. Thus, the proposed QHH does not require extra training time and could run efficiently using a single processor. This feature may broaden the possible applications of the proposed method in real-world scenarios.

I. Analysis of Algorithmic Complexity

Given its multi-component structure, the runtime of the QHH has become a focal point of our attention. The most computationally intensive component is the action execution, including the following steps.

Step 1: Adjust the solution with LLH for $\mu \cdot \xi$ times. The complexity of this step is $O(n)$.

Step 2: Execute 2-opt with probability for $\mu \cdot \xi$ times. The complexity is $O(n^2)$ according to Algorithm 4.

Step 3: Repair the solution with GIRR for μ times. GIRR consists of two separate steps, insertion and removal. The insertion takes the complexity of $O(nm)$, while the removal costs $O(n^3)$. Thus, the complexity of step 3 is $O(nm + n^3)$.

Therefore, the time complexity of executing an action is $O(n^3 + n^2 + nm + n) = O(n^3)$, since $n \gg m$ in the real world. In these steps, the repetition of steps $\mu \cdot \xi$ and μ is ignored since they are constants. In many cases, the total number of generations is related to the size of the problem. For example, in a well-known benchmark [12], the total number of evaluations is 25000 times the number of nodes in the problem. In this case, the number of generations is $O(n)$. Therefore, the time complexity of the entire algorithm run is $O(n^4)$. Additionally, QHH facilitates online learning and single-point search. Thus, it does not require training time and can run efficiently on a single processor. This feature may broaden the possible applications of QHH in the real world.

V. Experiments

In this section, we perform computational simulations on 60 instances (with up to 1000 customers) in total. Among them, 17 are from a recent CEVRP benchmark suite designated for the IEEE-WCCI 2020 competition on Evolutionary Computation for the EVRP [12]. 24 are from another common CEVRP benchmark suite in the literature [51]. 19 are created based on the steps in [51]. Details of these instances are provided in part II of the supplementary materials. All algorithms are implemented using C++ and compiled with the GCC compiler. The experiments are performed on a server with Intel® Xeon® Gold 6330N CPU @ 2.20GHz and 128GiB memory, running an Ubuntu 20.04.5 LTS system. The source code and experimental data are accessible at a GitHub repository.[1]

A. Parameter Calibration

Four key parameters of QHH are calibrated: the increment of threshold δ, the probability of exploration ε, the decreasing coefficient κ, and the reconstruction factor Ω. Taguchi's method of design-of-experiment (DOE) is conducted on instance 13 to evaluate the influence of the parameters. The details of this experiment are provided in part III of the supplementary materials. As illustrated in Fig. 7, the best combination of parameters is $\delta = 15$, $\varepsilon = 0.04$, $\kappa = 0.98$, and $\Omega = 2000$. The remaining parameters of QHH are configured as outlined in Table II.

[1] https://github.com/Y1fanHE/qhh-for-cevrp

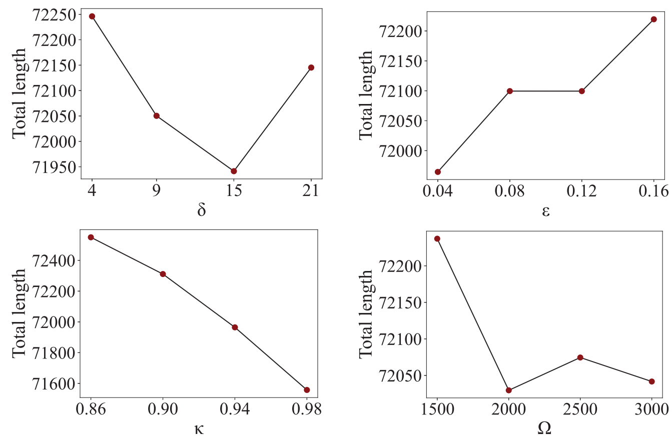


Fig. 7. Results of parameter calibration.

TABLE II
PARAMETER SETTINGS

Parameter	Definition	Value
α	Learning rate in Bellman equation	0.5
γ	Discount factor in Eq. (11)	1
Q_{init}	Initial value of Q-table	0.01
λ	Threshold discount coefficient	0.22
μ	Repetitions of GIRR in an action	3
ξ	Repetitions of a LLH in an action	40
η	Search depth	5
b	Increment amount of δ	5
P_{opt}	Probability of executing 2-opt	1 or 0.2
M	Parameter in reward function	0.05
ν	Generations to decrease λ	40
β	Reward bias	0.01 or -0.008

B. Comparison With the State-of-the-Art Algorithms

In this section, the QHH is compared with several state-of-the-art algorithms, namely threshold acceptance-based multi-layer search (TAMLS) [28], BACO [14], and confidence-based ant colony optimization (CACO) [26], with its two variants CACO-I and CACO-D using different encoding schemes. For a fair comparison, all five algorithms are executed on the same machine under the predefined execution time for 30 repetitions on each instance. The termination criteria are set to $\theta \cdot (|I| + |F|)/1000$ hours [28]. θ equals 1, 2, and 3 for the combined number of customers and distribution center falls in the ranges $[22, 101]$, $[102, 916]$, and $[917, 1001]$, respectively.

Table III provides the performance of the five compared algorithms across all 60 test instances in terms of the best and mean route lengths obtained from each algorithm. The visual representation of the data can be found in part IV of the supplementary materials. QHH outperforms the comparison algorithms on 55 of 60 instances (91.67%) in terms of best route lengths and 52 of 60 instances (86.67%) in terms of mean route lengths. Notably, QHH exhibits superior performance in large-scale instances with over 150 customers (i.e., instances 9–17 and 31–60), consistently outperforming all comparison algorithms in these complex scenarios. Additionally, a Wilcoxon signed-rank test at 95% confidence level with Bonferroni correction in Table IV shows the effectiveness of the QHH over all other methods is

TABLE III
THE RESULTS OBTAINED BY ALL COMPARED ALGORITHMS ON 60 CEVRP INSTANCES

No.	Time(h)	Best					Mean				
		TAMLS	BACO	CACO-I	CACO-D	QHH	TAMLS	BACO	CACO-I	CACO-D	QHH
1	0.030	**384.67**	**384.67**	**384.67**	**384.67**	**384.67**	**385.12**	**384.67**	**384.67**	**384.67**	**384.67**
2	0.032	**571.94**	**571.94**	**571.94**	**571.94**	**571.94**	**571.94**	**571.94**	**571.94**	**571.94**	**571.94**
3	0.036	**509.47**	**509.47**	**509.47**	**509.47**	**509.47**	509.56	**509.47**	**509.47**	**509.47**	522.50
4	0.039	**840.14**	840.56	840.56	840.56	**840.14**	840.24	841.95	842.24	840.56	**840.14**
5	0.060	**529.90**	**529.90**	**529.90**	**529.90**	**529.90**	542.65	530.41	**529.90**	**529.90**	**529.90**
6	0.085	694.54	**692.64**	**692.64**	**692.64**	**692.64**	714.14	693.03	693.43	**692.64**	692.89
7	0.110	855.87	837.89	840.25	838.83	**834.22**	879.88	845.24	843.53	840.41	**837.81**
8	0.294	16634.27	16064.74	15917.57	15903.53	**15897.24**	17514.75	16164.15	16005.88	**15946.53**	16036.62
9	0.446	11993.25	11199.52	11114.60	11610.88	**11078.27**	11993.25	11329.23	11230.00	11713.91	**11162.29**
10	0.772	27300.61	26594.68	26469.12	27753.89	**26409.67**	27322.69	26862.02	26637.70	27968.96	**26570.48**
11	0.958	26329.44	24900.99	24824.25	25802.71	**24561.65**	26329.44	25049.08	24969.84	25928.03	**24785.40**
12	1.158	53447.51	52349.36	53979.95	52831.60	**51551.36**	53447.51	53178.85	54625.80	52951.81	**51692.69**
13	1.420	71936.75	71726.16	71966.24	75998.30	**70916.83**	71947.72	72206.92	72572.22	76498.85	**71342.06**
14	1.558	80201.48	80437.08	80410.09	82985.97	**79322.23**	80201.48	80830.13	80828.41	83291.54	**79608.60**
15	1.688	166946.98	164933.60	165074.94	166187.29	**161881.70**	166948.77	165581.88	166360.50	166525.12	**162419.50**
16	1.850	344217.72	343984.51	345428.58	343724.07	**336292.36**	344217.72	345744.33	346616.98	344850.12	**336850.15**
17	3.030	78184.30	76842.26	78040.30	79412.98	**75373.23**	78193.33	78186.81	79029.72	79773.83	**75848.78**
18	0.029	**378.44**	**378.44**	**378.44**	**378.44**	**378.44**	**378.44**	**378.44**	**378.44**	**378.44**	**378.44**
19	0.030	**569.53**	**569.53**	**569.53**	**569.53**	**569.53**	**569.53**	**569.53**	**569.53**	**569.53**	**569.53**
20	0.035	**514.92**	**514.92**	**514.92**	**514.92**	**514.92**	515.33	**514.92**	**514.92**	**514.92**	528.18
21	0.037	**845.72**	**845.72**	**845.72**	**845.72**	**845.72**	846.44	847.35	846.19	**845.72**	**845.72**
22	0.049	**727.74**	**727.74**	**727.74**	**727.74**	**727.74**	727.80	**727.74**	**727.74**	**727.74**	**727.74**
23	0.060	528.77	528.77	528.77	528.77	**528.46**	541.12	529.30	528.77	528.77	**528.46**
24	0.080	254.84	**242.74**	**242.74**	**242.74**	**242.74**	254.92	243.63	242.97	**242.74**	**242.74**
25	0.089	697.58	**690.86**	**690.86**	**690.86**	**690.86**	713.66	695.64	692.69	691.76	**691.61**
26	0.112	848.27	834.99	834.16	834.97	**831.10**	869.00	839.66	836.74	835.77	**832.22**
27	0.110	**828.55**	**828.55**	**828.55**	**828.55**	**828.55**	833.59	**828.55**	**828.55**	**828.55**	**828.55**
28	0.126	1049.53	1046.91	1045.38	1045.15	**1045.14**	1062.73	1049.51	1046.16	**1045.24**	1069.39
29	0.140	1211.85	1165.70	**1163.33**	**1163.33**	1164.29	1216.71	1175.13	1167.86	**1164.07**	1166.04
30	0.294	16340.91	15932.59	15821.95	**15817.51**	15817.88	17269.99	16017.33	**15918.83**	15934.78	15921.49
31	0.326	1079.00	1037.65	**1032.77**	1045.50	1033.24	1120.09	1047.74	1042.00	1051.52	**1037.67**
32	0.424	1395.80	1323.17	**1309.88**	1371.37	1312.47	1398.12	1343.00	1326.21	1381.12	**1320.73**
33	0.442	12053.29	11179.94	11069.50	11429.35	**10999.49**	12053.29	11290.34	11194.38	11640.99	**11091.71**
34	0.720	27067.41	26526.39	26354.70	27628.16	**26279.09**	27067.41	26683.53	26544.13	27784.96	**26426.03**
35	0.938	26324.27	24823.24	24713.05	25692.88	**24577.68**	26324.27	25076.54	24908.39	25912.57	**24744.24**
36	1.154	53328.13	52221.92	54056.98	52654.28	**51406.36**	53328.13	52833.65	54464.47	52774.73	**51571.77**
37	1.396	71588.27	70483.07	70273.03	74370.56	**69477.45**	71589.17	70838.76	70908.32	74924.46	**69820.45**
38	1.518	79641.71	79567.62	79693.19	82194.55	**78586.04**	79641.71	80247.27	80201.84	82651.62	**78982.73**
39	1.660	166560.10	163785.54	164158.44	165040.39	**161072.76**	166560.10	164504.52	165227.01	165312.82	**161408.05**
40	1.840	342889.55	341416.42	341478.37	341290.74	**333499.82**	342889.55	342751.26	343442.46	341937.90	**334130.37**
41	3.017	77794.15	76160.24	77123.12	78864.53	**74785.39**	77800.59	77260.28	78360.98	79172.31	**75522.32**
42	0.428	45716.36	45014.61	44851.56	45512.86	**44467.21**	45718.73	45333.37	45119.24	45751.89	**44669.78**
43	0.584	22405.54	21017.52	20748.24	21414.72	**20715.45**	22405.54	21180.96	20932.30	21485.18	**20910.02**
44	0.622	36073.50	34750.97	34652.15	36443.54	**34444.39**	36090.06	35067.48	34924.32	36737.30	**34660.68**
45	0.786	68573.05	67310.59	67154.52	69352.19	**66623.45**	68573.05	67682.83	67524.53	69576.60	**66852.49**
46	0.826	40933.37	39120.42	39001.63	40380.05	**38882.19**	40933.37	39422.62	39207.56	40607.67	**38978.62**
47	0.842	21462.34	20547.86	**20237.30**	21301.22	20394.09	21462.34	20780.82	**20465.68**	21436.39	20569.88
48	0.882	112885.68	110241.18	109623.64	111295.15	**108465.34**	112885.68	110942.64	110098.77	111662.34	**108778.74**
49	1.114	89789.70	87410.47	87535.99	89260.78	**87184.96**	89789.70	87893.85	87786.84	89531.34	**87322.35**
50	1.168	45690.94	43960.77	43952.20	46005.71	**43375.07**	45690.94	44382.15	44342.12	46264.75	**43671.67**
51	1.276	62850.44	61806.61	61527.13	65099.23	**60633.45**	62850.44	62146.23	62099.18	65608.57	**61008.55**
52	1.264	65587.83	63685.38	63705.28	64933.81	**63398.62**	65587.83	63910.12	64067.08	65094.86	**63688.40**
53	1.332	108230.97	107070.90	107083.50	107768.68	**107004.24**	108230.97	107255.78	107223.37	107898.70	**107097.31**
54	1.358	158886.47	154680.45	154696.15	153256.47	**148037.51**	158887.10	156272.20	156965.95	153581.25	**148747.41**
55	1.416	85464.29	84393.59	84710.13	86804.62	**83535.72**	85464.29	84840.67	85184.29	87167.37	**83875.98**
56	1.584	76511.37	75012.54	75056.13	79032.69	**74230.18**	76516.61	75730.24	75948.60	79276.20	**74684.82**
57	1.618	77237.74	75042.11	75269.32	77032.20	**74486.35**	77238.08	75445.00	75859.21	77032.20	**75036.12**
58	1.760	92831.28	90290.82	90409.33	92610.95	**90283.30**	92831.28	90637.50	90732.33	92783.42	**90583.89**
59	1.920	146046.09	143410.11	143474.27	141919.47	**135191.83**	146046.59	144811.87	145629.46	142616.04	**136186.40**
60	1.968	124168.37	124439.23	124534.94	124513.66	**122345.50**	124177.32	125241.72	125172.72	124785.19	**123433.81**

statistically significant. Table V provides a pairwise comparison matrix among all six compared algorithms with respect to the mean route lengths across all 60 instances. The value in the i-th row and j-th column indicates the number of wins of the i-th algorithm compared to the j-th algorithm. As shown in the table, the best and the second-best algorithms are QHH (total wins =197) and CACO-I (total wins =118), respectively.

Fig. 8 illustrates the relative improvement in route length of QHH compared to the second-best algorithm CACO-I. The relative improvement on an instance is computed as $\Delta f = (f_{\text{CACO-I}} - f_{\text{QHH}})/f_{\text{CACO-I}}$, where f_{QHH} and $f_{\text{CACO-I}}$ are

TABLE IV
P-VALUES OF WILCOXON SIGNED-RANK TEST ON 60 CEVRP INSTANCES

Method	TAMLS	BACO	CACO-I	CACO-D
P-value	2.12E-10	1.38E-09	2.92E-08	9.14E-09

TABLE V
PAIRWISE COMPARISON MATRIX RESPECT TO MEAN ROUTE LENGTHS

Algorithm	TAMLS	BACO	CACO-I	CACO-D	QHH	Total
TAMLS	0	7	11	8	3	29
BACO	50	0	19	32	3	104
CACO-I	46	33	0	33	6	118
CACO-D	39	20	17	0	6	82
QHH	54	51	47	45	0	197

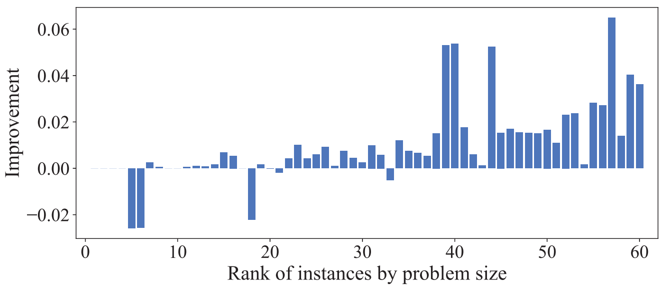
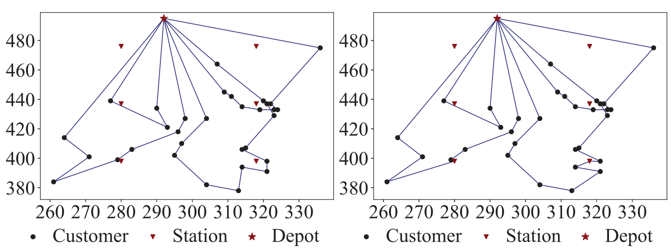


Fig. 8. Relative improvement of QHH compared to CACO-I.

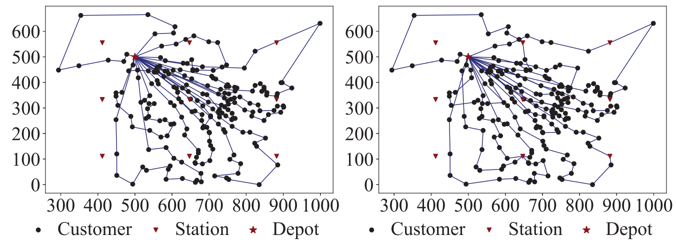


(a) CACO-I, total length=840.56 (b) QHH, total length=840.14

Fig. 9. The best solutions by CACO-I and QHH on instance 4.

the mean route length of QHH and CACO-I on that instance, respectively. The instances are arranged by their ranks in problem sizes. Obviously, QHH shows significant improvement especially on large-scale instances (over 6% improvement at maximum). Moreover, since the relationship between energy consumption and route length is linear in this research (as well as in many other studies [52]), improvements in route length directly translate into corresponding improvements in energy consumption. This highlights the energy-saving advantages of our proposed QHH approach.

Figs. 9 and 10 illustrate the best delivery routes generated by QHH and the second-best algorithm CACO-I on instances 4 and 9, respectively. For instance 4, the main difference is only in one of the routes; however, for instance 9, the allocation of customers to the routes is very different between the best solutions by CACO-I and QHH. This may indicate that QHH has a stronger global search ability compared to CACO-I on large-scale instances.



(a) CACO-I, total length=11114.60 (b) QHH, total length=11078.27

Fig. 10. The best solutions by CACO-I and QHH on instance 9.

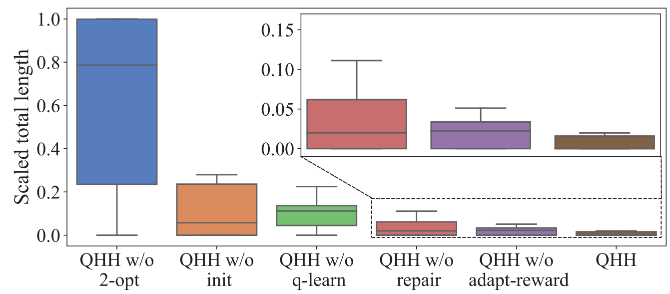


Fig. 11. Scaled total lengths of the solutions by QHH and its ablations.

C. Ablation Experiments

Several components are proposed or employed to enhance the performance of the QHH. Therefore, an ablation experiment is conducted to investigate their effectiveness. Five ablations are created by either removing these components or replacing them with their standard counterparts.

1) **QHH w/o 2-opt**: This ablation eliminates the steps of 2-opt local search inside the execution of actions.
2) **QHH w/o init**: This ablation employs a common initialization rather than the proposed one.
3) **QHH w/o q-learn**: This ablation randomly selects actions instead of using the Q-table.
4) **QHH w/o repair**: This ablation uses the repair method in BACO instead of the proposed repair method.
5) **QHH w/o adapt-reward**: This ablation eliminates the adaptive (time-dependent) term $\max\left\{M, \frac{t}{T}\right\}$ in the reward function.

Fig. 11 illustrates the min-max scaled total lengths of the delivery routes found by QHH and its ablations on instances 1–17 [12] in 20 repetitions. As illustrated in the figure, QHH outperforms all ablations. Therefore, this observation verifies the effectiveness of the five components.

In QHH, a set of LLHs serves as the basic components of the algorithm, increasing the diversity of search patterns. Similar ideas are found in variable neighborhood search (VNS), where operators are executed in a predefined order, and multi-neighborhood search (MNS), where operators are selected randomly. However, QHH manages LLHs using Q-learning, which allows LLHs to be executed based on their contributions to the search history and benefits the search in two ways. First, compared to MNS, the probability of using different LLHs is determined by their estimated quality based on the search history. Second, compared to VNS, the order of using LLHs is not fixed but determined by the agent's state. Therefore, QHH integrates three strengths: effective local search, diverse search patterns, and intelligent management of LLHs, and consequently solves the problem efficiently.

VI. Conclusion

In this paper, a QHH algorithm was proposed to solve the CEVRP. A novel initialization strategy was presented to construct the capacity-feasible route. An improved repair method for inserting charging stations into a capacity-feasible route was developed. Six common heuristics were employed to construct the LLH set, which serves as the available set of actions. The Q-learning algorithm was utilized as the HLH to select the most suitable action and refine the solutions. A new reward-penalty strategy was proposed to facilitate an efficient learning process. The parameter configurations of QHH were explored using the DOE method, and the performance of QHH was assessed using a large set of benchmark instances. Comparative analyses with four state-of-the-art algorithms from the literature underscored the superiority of the proposed QHH algorithm.

In the future, we plan to broaden the application scope of QHH into a wide range of VRPs, including but not limited to the multi-depot VRP [53], open VRP [54], and VRP with time windows [55]. Furthermore, we aim to enhance the interpretability of the Q-learning by incorporating explainable machine learning models or utilizing visualization techniques. These enhancements will provide more transparent insights into the decision-making processes, thereby improving the performance and applicability. Additionally, we would like to consider other effective initialization and heuristics, such as the Kuhn-Munkres method, to improve the QHH.

Acknowledgment

The authors would like to thank the anonymous reviewers for their valuable comments and suggestions.

References

[1] Y. V. Fan, S. Perry, J. J. Klemeš, and C. T. Lee, "A review on air emissions assessment: Transportation," *J. Cleaner Prod.*, vol. 194, pp. 673–684, Sep. 2018.

[2] B. Golden, X. Wang, and E. Wasil, "The evolution of the vehicle routing problem—A survey of VRP research and practice from 2005 to 2022," in *The Evolution of the Vehicle Routing Problem: A Survey of VRP Research and Practice From 2005 to 2022*. Cham, Switzerland: Springer, 2023, pp. 1–64.

[3] F. Un-Noor, S. Padmanaban, L. Mihet-Popa, M. Mollah, and E. Hossain, "A comprehensive study of key electric vehicle (EV) components, technologies, challenges, impacts, and future direction of development," *Energies*, vol. 10, no. 8, p. 1217, Aug. 2017.

[4] D. O'Rourke, "Can sustaining innovation deliver sustainability? Amazon's innovation processes on corporate decarbonization," *California Manage. Rev.*, vol. 67, pp. 58–81, Jan. 2025.

[5] S. N. Kumar and R. Panneerselvam, "A survey on the vehicle routing problem and its variants," *Intell. Inf. Manage.*, vol. 4, no. 3, pp. 66–74, 2012.

[6] G. D. Konstantakopoulos, S. P. Gayialis, and E. P. Kechagias, "Vehicle routing problem and related algorithms for logistics distribution: A literature review and classification," *Oper. Res.*, vol. 22, no. 3, pp. 2033–2062, Jul. 2022.

[7] S. Öztürk and A. E. Kuzucuoğlu, "Optimal bid valuation using path finding for multi-robot task allocation," *J. Intell. Manuf.*, vol. 26, no. 5, pp. 1049–1062, Oct. 2015.

[8] S. Öztürk and A. E. Kuzucuoğlu, "A multi-robot coordination approach for autonomous runway foreign object debris (FOD) clearance," *Robot. Auto. Syst.*, vol. 75, pp. 244–259, Jan. 2016.

[9] A. M. Altabeeb, A. M. Mohsen, L. Abualigah, and A. Ghallab, "Solving capacitated vehicle routing problem using cooperative firefly algorithm," *Appl. Soft Comput.*, vol. 108, Sep. 2021, Art. no. 107403.

[10] M. Amous, S. Toumi, B. Jarboui, and M. Eddaly, "A variable neighborhood search algorithm for the capacitated vehicle routing problem," *Electron. Notes Discrete Math.*, vol. 58, pp. 231–238, Apr. 2017.

[11] J. Lin, W. Zhou, and O. Wolfson, "Electric vehicle routing problem," *Transp. Res. Proc.*, vol. 12, pp. 508–521, Jan. 2016.

[12] M. Mavrovouniotis, C. Menelaou, S. Timotheou, C. Panayiotou, G. Ellinas, and M. Polycarpou, "Benchmark set for the IEEE WCCI-2020 competition on evolutionary computation for the electric vehicle routing problem," *KIOS CO*, Tech. Rep., 2020.

[13] B. Vahedi-Nouri, H. Arbabi, F. Jolai, R. Tavakkoli-Moghaddam, and A. Bozorgi-Amiri, "Bi-objective collaborative electric vehicle routing problem: Mathematical modeling and matheuristic approach," *J. Ambient Intell. Humanized Comput.*, vol. 14, no. 8, pp. 10277–10297, Aug. 2023.

[14] Y.-H. Jia, Y. Mei, and M. Zhang, "A bilevel ant colony optimization algorithm for capacitated electric vehicle routing problem," *IEEE Trans. Cybern.*, vol. 52, no. 10, pp. 10855–10868, Oct. 2022.

[15] F. Zahedi, H. Kia, and M. Khalilzadeh, "A hybrid metaheuristic approach for solving a bi-objective capacitated electric vehicle routing problem with time windows and partial recharging," *J. Adv. Manage. Res.*, vol. 20, no. 4, pp. 695–729, Oct. 2023.

[16] S. Pourazarm, C. G. Cassandras, and A. Malikopoulos, "Optimal routing of electric vehicles in networks with charging nodes: A dynamic programming approach," in *Proc. IEEE Int. Electric Vehicle Conf. (IEVC)*, Dec. 2014, pp. 1–7.

[17] E. N. Duman, D. Taş, and B. Çatay, "Branch-and-price-and-cut methods for the electric vehicle routing problem with time windows," *Int. J. Prod. Res.*, vol. 60, no. 17, pp. 5332–5353, Sep. 2022.

[18] X. Zuo, Y. Xiao, C. Zhu, and M. You, "A linear MIP model for the electric vehicle routing problem with time windows considering linear charging," in *Proc. Annu. Rel. Maintainability Symp. (RAMS)*, Jan. 2018, pp. 1–5.

[19] V. Q. Hien, T. C. Dao, and H. T. T. Binh, "A greedy search based evolutionary algorithm for electric vehicle routing problem," *Int. J. Speech Technol.*, vol. 53, no. 3, pp. 2908–2922, Feb. 2023.

[20] M. E. H. Sadati and B. Çatay, "A hybrid variable neighborhood search approach for the multi-depot green vehicle routing problem," *Transp. Res. E, Logistics Transp. Rev.*, vol. 149, May 2021, Art. no. 102293.

[21] E. A. Silver, "An overview of heuristic solution methods," *J. Oper. Res. Soc.*, vol. 55, no. 9, pp. 936–956, Sep. 2004.

[22] L. Cooper, "Heuristic methods for location-allocation problems," *SIAM Rev.*, vol. 6, no. 1, pp. 37–53, Jan. 1964.

[23] D. Woller, V. Kozák, and M. Kulich, "The GRASP metaheuristic for the electric vehicle routing problem," in *Proc. Int. Conf. Modeling Simulation Auto. Syst.*, 2021, pp. 189–205.

[24] C. Wang, C. Guo, and X. Zuo, "Solving multi-depot electric vehicle scheduling problem by column generation and genetic algorithm," *Appl. Soft Comput.*, vol. 112, Nov. 2021, Art. no. 107774.

[25] M. R. C. O. Leite, H. S. Bernardino, and L. B. Gonçalves, "A variable neighborhood descent with ant colony optimization to solve a bilevel problem with station location and vehicle routing," *Int. J. Speech Technol.*, vol. 52, no. 7, pp. 7070–7090, May 2022.

[26] Y.-H. Jia, Y. Mei, and M. Zhang, "Confidence-based ant colony optimization for capacitated electric vehicle routing problem with comparison of different encoding schemes," *IEEE Trans. Evol. Comput.*, vol. 26, no. 6, pp. 1394–1408, Dec. 2022.

[27] V. D. Bui and T. Mai, "Imitation improvement learning for large-scale capacitated vehicle routing problems," in *Proc. Int. Conf. Automated Planning Scheduling*, 2023, vol. 33, no. 1, pp. 551–559.

[28] Y. Chen, J. Xue, Y. Zhou, and Q. Wu, "An efficient threshold acceptance-based multi-layer search algorithm for capacitated electric vehicle routing problem," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 6, pp. 5867–5879, Jun. 2024.

[29] M. Pustilnik and F. Borrelli, "Clustering heuristics for robust energy capacitated vehicle routing problem (ECVRP)," 2024, *arXiv:2403.13906*.

[30] S. P. Adam, S.-A. N. Alexandropoulos, P. M. Pardalos, and M. N. Vrahatis, "No free lunch theorem: A review," in *Approximation and Optimization*. Cham, Switzerland: Springer, 2019, pp. 57–82, doi: 10.1007/978-3-030-12767-1_5.

[31] E. Soubeiga, "Development and application of hyperheuristics to personnel scheduling," Ph.D. dissertation, School Comput. Sci. Inf. Technol., Univ. Nottingham, Nottingham, U.K., 2003.

[32] E. K. Burke et al., "Hyper-heuristics: A survey of the state of the art," *J. Oper. Res. Soc.*, vol. 64, no. 12, pp. 1695–1724, Dec. 2013.

[33] J. H. Drake, A. Kheiri, E. Özcan, and E. K. Burke, "Recent advances in selection hyper-heuristics," *Eur. J. Oper. Res.*, vol. 285, no. 2, pp. 405–428, Sep. 2020.

[34] G. L. Pappa, G. Ochoa, M. R. Hyde, A. A. Freitas, J. Woodward, and J. Swan, "Contrasting meta-learning and hyper-heuristic research: The role of evolutionary algorithms," *Genetic Program. Evolvable Mach.*, vol. 15, no. 1, pp. 3–35, Mar. 2014.

[35] H.-B. Song, Y.-H. Yang, J. Lin, and J.-X. Ye, "An effective hyper heuristic-based memetic algorithm for the distributed assembly permutation flow-shop scheduling problem," *Appl. Soft Comput.*, vol. 135, Mar. 2023, Art. no. 110022.

[36] Z. A. Aziz, "Ant colony hyper-heuristics for travelling salesman problem," *Proc. Comput. Sci.*, vol. 76, pp. 534–538, Jan. 2015.

[37] J. M. Balera and V. A. D. Santiago Júnior, "A systematic mapping addressing hyper-heuristics within search-based software testing," *Inf. Softw. Technol.*, vol. 114, pp. 176–189, Oct. 2019.

[38] R. Raghavjee and N. Pillay, "A genetic algorithm selection perturbative hyper-heuristic for solving the school timetabling problem," *ORiON*, vol. 31, no. 1, pp. 39–60, Apr. 2015.

[39] K. Z. Zamli, B. Y. Alkazemi, and G. Kendall, "A Tabu search hyper-heuristic strategy for t-way test suite generation," *Appl. Soft Comput.*, vol. 44, pp. 57–74, Jul. 2016.

[40] V. A. D. Santiago Júnior, E. Özcan, and V. R. D. Carvalho, "Hyper-heuristics based on reinforcement learning, balanced heuristic selection and group decision acceptance," *Appl. Soft Comput.*, vol. 97, Dec. 2020, Art. no. 106760.

[41] A. Steenson, E. Ozcan, A. Kheiri, B. McCollum, and P. McMullan, "An online learning selection hyper-heuristic for educational timetabling," in *Proc. 13th Int. Conf. Pract. Theory Automated Timetabling-PATAT*, vol. 1, 2021, pp. 129–146.

[42] J. Lin, Y.-Y. Li, and H.-B. Song, "Semiconductor final testing scheduling using Q-learning based hyper-heuristic," *Expert Syst. Appl.*, vol. 187, Jan. 2022, Art. no. 115978.

[43] M. Ghavamzadeh, H. J. Kappen, M. G. Azar, and R. Munos, "Speedy Q-learning," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 24, 2011, pp. 2411–2419.

[44] E. S. Low, P. Ong, and C. Y. Low, "A modified Q-learning path planning approach using distortion concept and optimization in dynamic environment for autonomous mobile robot," *Comput. Ind. Eng.*, vol. 181, Jul. 2023, Art. no. 109338.

[45] Z. J. Viharos and R. Jakab, "Reinforcement learning for statistical process control in manufacturing," *Measurement*, vol. 182, Sep. 2021, Art. no. 109616.

[46] Q. Chen, Q. Zhang, and Y. Liu, "Balancing exploration and exploitation in episodic reinforcement learning," *Expert Syst. Appl.*, vol. 231, Nov. 2023, Art. no. 120801.

[47] P. Winker, J. Chen, and D. K. J. Lin, "The construction of optimal design for order-of-addition experiment via threshold accepting," in *Contemporary Experimental Design, Multivariate Analysis and Data Mining*. Cham, Switzerland: Springer, 2020, pp. 93–109.

[48] J. Clifton and E. B. Laber, "Q-learning: Theory and applications," *Annu. Rev. Statist. Its Appl.*, vol. 7, no. 1, pp. 279–301, 2020.

[49] S. Lin, "Computer solutions of the traveling salesman problem," *Bell Syst. Tech. J.*, vol. 44, no. 10, pp. 2245–2269, Dec. 1965.

[50] K.-S. Hwang, H.-Y. Lin, Y.-P. Hsu, and H.-H. Yu, "Self-organizing state aggregation for architecture design of Q-learning," *Inf. Sci.*, vol. 181, no. 13, pp. 2813–2822, Jul. 2011.

[51] M. Mavrovouniotis, C. Menelaou, S. Timotheou, G. Ellinas, C. Panayiotou, and M. Polycarpou, "A benchmark test suite for the electric capacitated vehicle routing problem," in *Proc. IEEE Congr. Evol. Comput. (CEC)*, Jul. 2020, pp. 1–8.

[52] I. Kucukoglu, R. Dewil, and D. Cattrysse, "The electric vehicle routing problem and its variations: A literature review," *Comput. Ind. Eng.*, vol. 161, Nov. 2021, Art. no. 107650.

[53] J. Li, B. Tian Dai, Y. Niu, J. Xiao, and Y. Wu, "Multi-type attention for solving multi-depot vehicle routing problems," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 11, pp. 17831–17840, Nov. 2024.

[54] V. C. Soares and M. Roboredo, "On the exact solution of the multi-depot open vehicle routing problem," *Optim. Lett.*, vol. 18, no. 4, pp. 1053–1069, May 2024.

[55] A. Rave and P. Fontaine, "The location routing problem with time windows and load-dependent travel times for cargo bikes," *Eur. J. Oper. Res.*, vol. 323, no. 1, pp. 97–107, May 2025.

Jian Lin (Member, IEEE) received the Ph.D. degree in control theory and control engineering from Zhejiang University, Hangzhou, China, in 2013. He is currently a Professor with the School of Information Technology and Artificial Intelligence, Zhejiang University of Finance and Economics, Hangzhou. He had published more than 40 refereed articles. His research interests include computational intelligence, vehicle routing, production planning and scheduling, and machine learning.

Xintao Wang is currently pursuing the bachelor's degree in artificial intelligence with Zhejiang University of Finance and Economics, Hangzhou, China. His main research interests include evolutionary computation, hyper-heuristics, and their applications in vehicle routing problems.

Rui Niu is currently pursuing the bachelor's degree in artificial intelligence with Zhejiang University of Finance and Economics, Hangzhou, China. Her main research interests include vehicle routing and production scheduling.

Yifan He (Member, IEEE) received the Ph.D. degree in computer science from the University of Tsukuba, Tsukuba, Japan, in 2023. He is currently a Lecturer with the School of Information Technology and Artificial Intelligence, Zhejiang University of Finance and Economics, China. His research interests include evolutionary computation, machine learning, and artificial life, as well as their applications in combinatorial optimization such as vehicle routing and production scheduling.